

Billions, Anonymous

Bitcoin's next leap without a fork

Bitcoin 2025 – Las Vegas
5.28.2025

Paul Sztorc
CEO, LayerTwo Labs
Author, BIPs 300/301

This Talk – 20 mins

1. Scaling Bitcoin... to 8 billion users.
2. Strong Privacy... offered to all.
3. Accomplishing this, today, without a ~~softfork~~

pull request to Bitcoin Core.

-op return

-Ordinals 2023

A slide from Pleb-Fi 2023

A Bold Claim

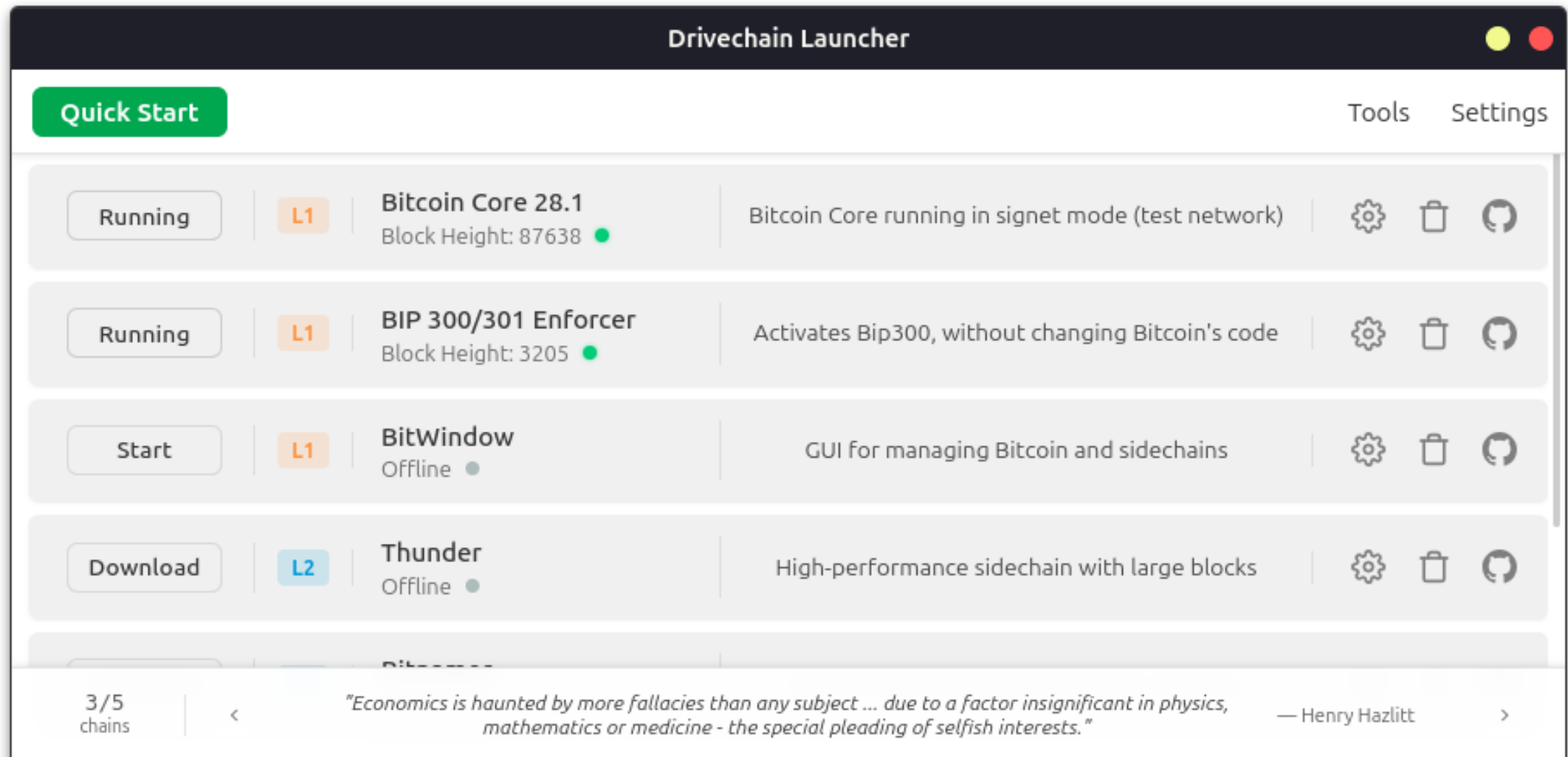
- BIP300 Solves All of Bitcoin's Biggest Problems
 - A. Heterogeneity Problem
 - B. Scalability
 - C. Privacy
 - D. Scams – Eliminating ScamCoins ; Domesticating the Token Casino
 - E. Security Budget
 - F. Decentralization
 - G. “Fundamental Value” of Bitcoin
- With...
 - H. ...zero risk to Bitcoin!

Endorsements

Visit www.LayerTwoLabs.com/friends for 47 more!

- "Drivechains...are pretty cool...and arguably could have been more important or useful than let's say Taproot."
 - **Adam Back**, Baltic Honeybadger 2022, Live on stage in front of everyone
- "We need Drivechain or all the work of thousands in the last 13 years will be in vain." ... "Drivechain is our only hope".
 - **fiatjaf**, (creator of nostr), on twitter
- "We should activate BIP300 and then ossify the baselayer".
 - **Robin Linus**, (creator of BitVM and ZeroSync), on twitter
- "We need your project, of course, for the obvious reasons..."
 - **Rene Pickhardt** (Author of *Mastering Lightning* , #1 stackoverflow (?) contributor for LN questions), MIT Bitcoin Expo, 2023

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com/Download

Drivechain Launcher

Quick Start

Tools

Settings

Running

L1

Bitcoin Core 28.1

Block Height: 87638

Bitcoin Core running in signet mode (test network)

Running

L1

BIP 300/301 Enforcer

Block Height: 3205

Activates Bip300, without changing Bitcoin's code

Start

L1

BitWindow

Offline

GUI for managing Bitcoin and sidechains

Download

L2

Thunder

Offline

High-performance sidechain with large blocks

3/5 chains

"Economics is haunted by more fallacies than any subject ... due to a factor insignificant in physics, mathematics or medicine - the special pleading of selfish interests."

— Henry Hazlitt

Parent Chain

Coins

BitNames

My Paymail

Messaging

Activity

Console / Logs

Transfer & Receive

Transaction Builder

My BitNames

Spend UTXO

Total: 5.12324678 BTC

kind	outpoint	value
regular	4306b2d6:1	0 BTC
regular	d5f7dc77:0	0.12345678 BTC
regular	ec1a4b91:1	4.99979000 BTC

Value In

Total: 0 BTC

kind	outpoint	value
------	----------	-------

Value Out

Total: 0 BTC

vout	address	va
------	---------	----

Balance: 85.12324678

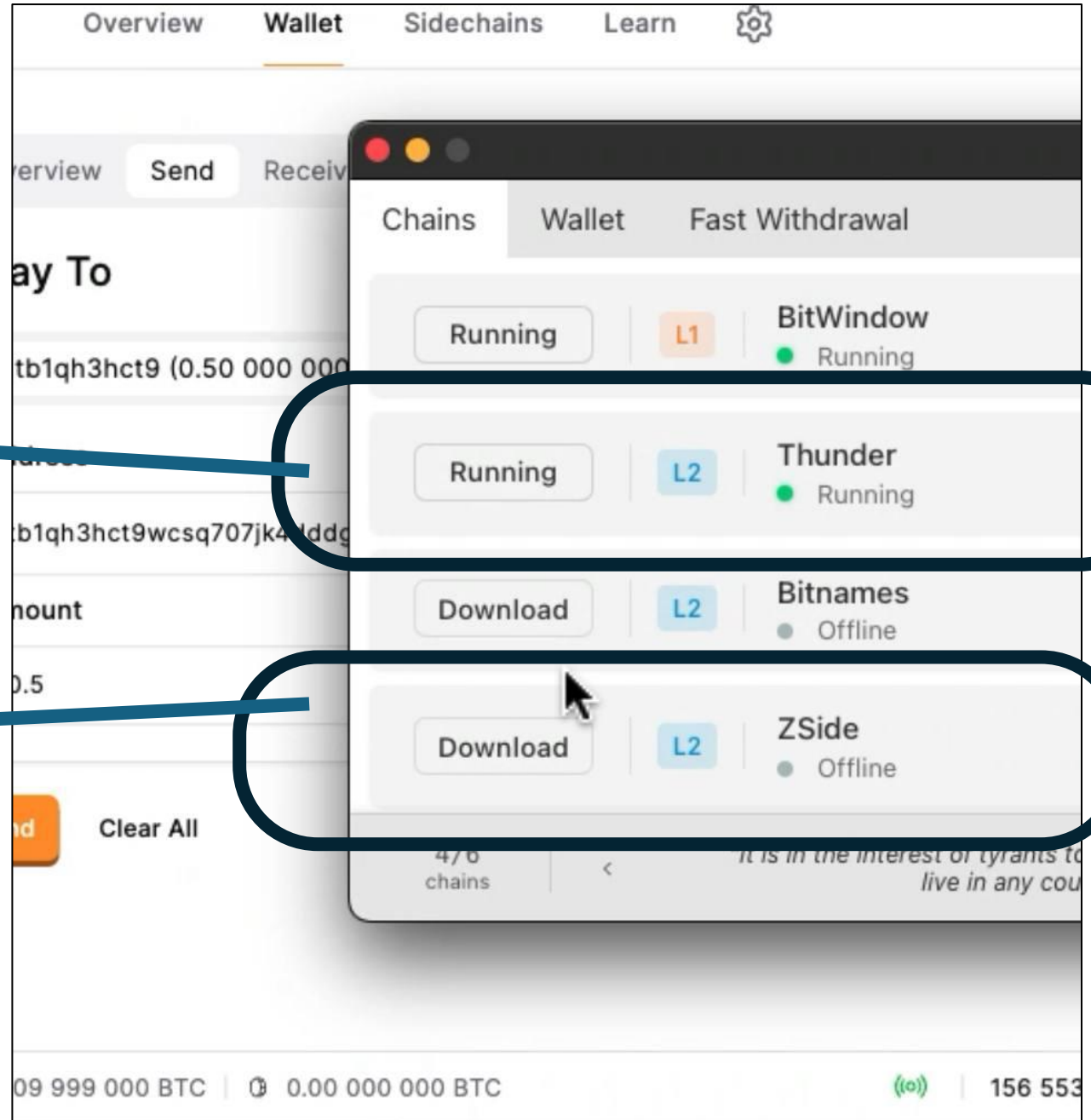
Block height: 3973

Best hash: 12be9c12...

Mine / Refresh

- Scaling to 8 billion [1/14]

- Privacy
 - Hiding sender
 - Hiding recipient
 - Hiding amount





Warning

This is experimental sidechain software. Use at your own risk!

Coin News

Stay up-to-date on the latest world developments

UW

SPECIAL PUBLIC ANNOUNCEMENT

20.may

ALL YOUR NEWS BELONGS TO THE PEOPLE

UW

Luke Dashjr's Stolen Bitcoin

7.may

Bitcoin Core developer Luke Dashjr is claiming to troll Dashjr with a cat

UW

Bitcoin Crashes to \$0. No one

7.may

in a plot twist that surprised about 100 million people clutching their Ledger wallets

Daemon Status

You can use BitWindow without the enforcer, but it's not that interesting because the wallet does not work.

Bitcoin Core (Patched) daemon

169 612 sync height

[View logs](#)

BIP300301 Enforcer daemon



1 535 / 169 612

0.91%

[View logs](#)

BitWindow daemon

169 612 sync height

[View logs](#)[< Previous](#)[1](#)[2](#)[3](#)[... 5](#)[> Next](#)[3 / page](#)

Latest Transactions

View the latest transactions on L1

Latest Blocks

View the latest blocks on the blockchain

0.00 000 000 BTC



1 535 / 169 612

0.91%

Last block: 43 seconds ago

5 peers

Plain Bitnames

ames My Paymail **Messaging** Activity Console / Logs

Encrypt ☒ Decrypt

Receiver's BitName or Encryption Pubkey (Bech32m):

04b10d894ca37b089de84792bd8e2bbbdd81130f

Ciphertext message (hex):

117686c58ff80a6b94b556efc021c2a86e92e72d1
e56ea44936bf25148ca366313f0ddae391067ab7a
0df4fb7cfdd86da4c4d346aa8936435e36791e262
3ae58c0e84e8e4c03f9d97c3656aa92cf390d2279
3c7b7e2a0880cf335762ab0db7eeff7ae79b6dbf8
1307c715a89afe76e9311a5a98b5e8bcb9e3f0a0e
b5daf75cac57d3cb7dea103f279bde

Decrypted message:

5468697320636f6d70757465722077696c6c2
073656c662d646573747275637420696e2031
30207365636f6e6473



Decrypted message (UTF-8):

This computer will self-destruct in
10 seconds

assets — bitcoind-signet — 80x24

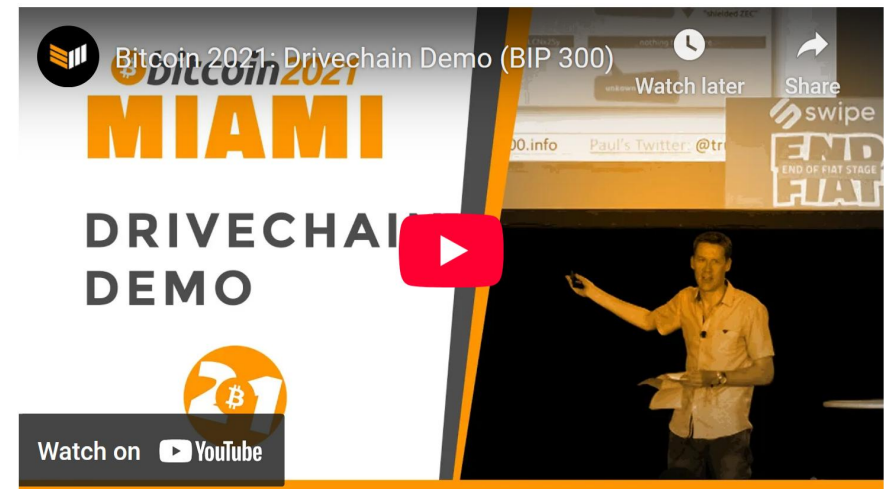
```
ffe941608cadff877c70a35c1 height=23375 version=0x20000000 log2_work=36.718845 tx
=23586 date='2025-01-30T01:56:12Z' progress=1.000000 cache=1.3MiB(10170txo)
2025-05-22T05:31:19Z UpdateTip: new best=0000006cd727e0970823f40c7ccf86b20132cc4
a0b59c98701640ec6300fe767 height=23376 version=0x20000000 log2_work=36.718907 tx
=23587 date='2025-01-30T01:57:12Z' progress=1.000000 cache=1.3MiB(10171txo)
2025-05-22T05:31:19Z UpdateTip: new best=0000016739265320145ddd8020c16f5d77d13d5
948752fb8538cd09295d0c531 height=23377 version=0x20000000 log2_work=36.718968 tx
=23588 date='2025-01-30T01:58:12Z' progress=1.000000 cache=1.3MiB(10172txo)
2025-05-22T05:31:19Z UpdateTip: new best=000000c626ead9153d7c0ca61661f496ead1902
9b91666532dc082045232d322 height=23378 version=0x20000000 log2_work=36.719030 tx
=23589 date='2025-01-30T01:59:12Z' progress=1.000000 cache=1.3MiB(10173txo)
2025-05-22T05:31:19Z UpdateTip: new best=000003754176418103f65fc037297d782d168b0
34686a8cfc186107dd76f2a0a height=23379 version=0x20000000 log2_work=36.719092 tx
=23590 date='2025-01-30T02:00:12Z' progress=1.000000 cache=1.3MiB(10174txo)
2025-05-22T05:31:19Z UpdateTip: new best=000001c148cd75982d3be0f839653ffc6928884
c1004246c994e9925e1b97d42 height=23380 version=0x20000000 log2_work=36.719154 tx
=23591 date='2025-01-30T02:01:12Z' progress=1.000000 cache=1.3MiB(10175txo)
2025-05-22T05:31:19Z UpdateTip: new best=0000032ec6c627f94554911eb70d98109d4a13c
6a584942f9d483140a048ffa3 height=23381 version=0x20000000 log2_work=36.719215 tx
=23592 date='2025-01-30T02:02:12Z' progress=1.000000 cache=1.3MiB(10176txo)
2025-05-22T05:31:19Z UpdateTip: new best=000002510b39129dd45ab86bc914de42ea65d61
bee82d8bb88e49b50735653b3 height=23382 version=0x20000000 log2_work=36.719277 tx
=23593 date='2025-01-30T02:03:12Z' progress=1.000000 cache=1.3MiB(10177txo)
```

bip300301-enforcer-latest-x86_64-apple-darwin — bip300301-enforcer-la...

```
:215: Stored 5 block header(s) in 66.213µs
2025-05-22T05:31:19.157535Z DEBUG initial_sync:sync_headers: bip300301_enforcer_
lib::validator::task: lib/validator/task/mod.rs:1079: Fetching batch of headers
starting from #23363 `00000d1ab6d843dcb5111418447ded28ac0c555402f8679d0cb34d1ea
7d19ed`
2025-05-22T05:31:19.158611Z DEBUG initial_sync:sync_headers:get_block_headers: b
ip300301_enforcer_lib::validator::main_rest_client: lib/validator/main_rest_clie
nt.rs:124: Fetched 4 block header(s) in 1.01327ms: 23363 -> 23366 block_hash=000
000d1ab6d843dcb5111418447ded28ac0c555402f8679d0cb34d1ea7d19ed descendants=2000
2025-05-22T05:31:19.158787Z DEBUG initial_sync:sync_headers:put_headers: bip3003
01_enforcer_lib::validator::dbs::block_hashes: lib/validator/dbs/block_hashes.rs
:215: Stored 4 block header(s) in 98.783µs
2025-05-22T05:31:19.196509Z DEBUG initial_sync:sync_headers: bip300301_enforcer_
lib::validator::task: lib/validator/task/mod.rs:1079: Fetching batch of headers
starting from #23366 `0000037594aaf973638a26b300c29a0dd7d648670deebb8ad609dc6a74
7be090`
2025-05-22T05:31:19.197667Z DEBUG initial_sync:sync_headers:get_block_headers: b
ip300301_enforcer_lib::validator::main_rest_client: lib/validator/main_rest_clie
nt.rs:124: Fetched 4 block header(s) in 1.068696ms: 23366 -> 23369 block_hash=00
00037594aaf973638a26b300c29a0dd7d648670deebb8ad609dc6a747be090 descendants=2000
2025-05-22T05:31:19.197850Z DEBUG initial_sync:sync_headers:put_headers: bip3003
01_enforcer_lib::validator::dbs::block_hashes: lib/validator/dbs/block_hashes.rs
:215: Stored 4 block header(s) in 69.32µs
2025-05-22T05:31:19.235433Z DEBUG initial_sync:sync_headers: bip300301_enforcer_
```

Long History [Explanations & Demos]

- Scaling
 - Nov 2015 – Drivechain – (31 pages)
 - Oct 2016 – Scaling III Milan – Presentation
 - Mar 2018 / Feb 2021 / May 2024 -- (36 pages) -- Detailed examples & numbers
- Privacy
 - Apr 2020 – L2 contest
 - Mar 2021 – software & demo video
- Overall
 - June 2021 – BitcoinConf
 - Baltic Honeybadger, TabConf, etc





Bitcoin 2021: Drivechain Demo (BIP 300)



Watch later



Share

MIAMI

**DRIVECHAIN
DEMO**



Watch on  YouTube



Critics ...

... we do have them, of course.

Peter Todd

Drivechains: A Detailed Analysis

Oct 13, 2023

Drivechains is a controversial proposal aiming to allow for the creation of sidechains containing coins meant to be pegged 1:1 to Bitcoin. Here we will analyze what drivechains actually proposes, technical flaws of drivechains, how drivechains could affect Bitcoin itself, and finally, discuss whether or not the idea is actually likely to see significant usage.

Credit goes to Matthew Haywood in particular for productive discussions about the characteristics and flaws of drivechains. He has published his own critique of drivechains [here](#).

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4. [Hashrate Escrows](#)
 - 4.1. [BIP-300: The Drivechain Take on Hashrate Escrows](#)
 - 4.1.1. [Other Unimportant BIP-300 Criticism](#)
5. [BIP-301: Blind Merge Mining](#)
 - 5.1. [Equivocation Attack](#)
 - 5.2. [Reorgs Invalidating Coins](#)

drivechain.info/peer-review/
peter-todd-2023.png

https://peterodd.org/2023/drivechains

Peter Todd

Drivechains: A Detailed Analysis

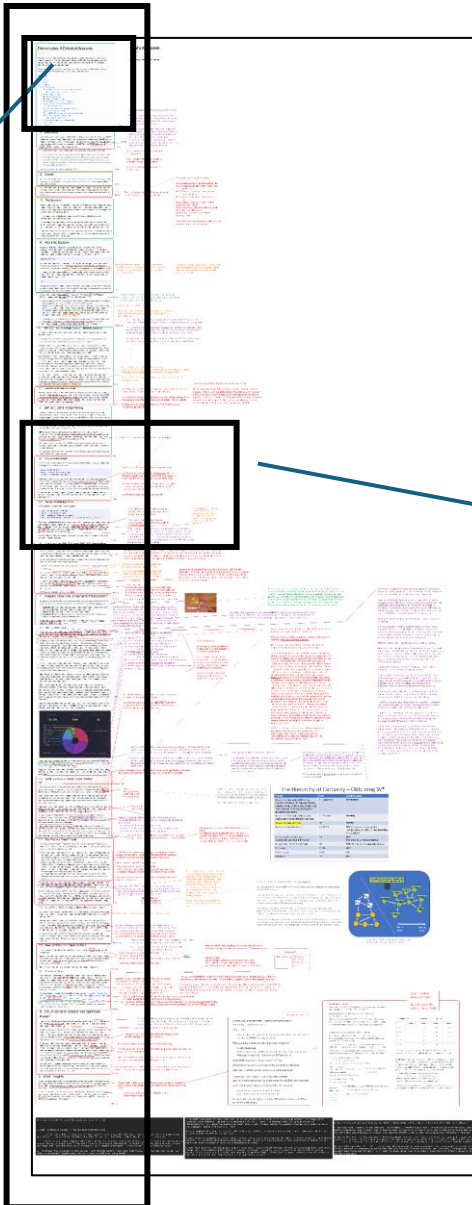
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5. BIP-301: Blind Merge Mining
 - 5.1. Equivocation Attack
 - 5.2. Reorgs Invalidating Coins



BIP-301 doesn't actually specify in detail how a blind-mined chain is meant to actually form a consensus. Those specific plans would be part of the blind-mined chain's consensus, not Bitcoin itself. However it's notable that for the most obvious and efficient way of implementing those rules, BIP-301 has a serious cryptographic flaw which we will discuss separately.

With respect to Bitcoin, you can think of BIP-301 as a mechanism where mining a special type of transaction requires a miner to also add a specific 'OP_RETURN' output to their coinbase transaction that commits to 32-bytes.

Since only one such transaction can be included, the economically rational outcome of the mechanism is a fee-based auction: highest fee wins.

This is just a mistake made by Peter. As I will discuss below.

True

5.1 Equivocation Attack

As per BIP-301, the result of an accepted Blind Merge Mined block is that an 'OP_RETURN' output in the following format is added to the coinbase:

- 1-byte - OP_RETURN (0x4a)
- 4-bytes - Message header (0x01617368)
- 32-bytes - h* (obtained from Simon)

An obvious problem with this design is it's subject to equivocation attacks. The obvious and efficient implementation of a merge mined chain is to use a SPV proof to the coinbase transaction to prove that a block has been blind-merge-mined. But since the OP_RETURN format doesn't commit to which chain is being blind-merge-mined, a miner could in fact include multiple conflicting OP_RETURN transactions at the same block height. This is similar to the 'double-proof' problem that Namecoin solved back in 2011.

An obvious solution would be to tag the OP_RETURN outputs with some kind of merge-mined chain identifier, and implement a strict rule consensus rule in the blind-merge-mined chain where the first matching tag wins.

This is a mistaken assumption about how Bip301 works.

Bip301 assumes that the L2 chain has full access to L1. So there is no need to use an SPV proof to the L1 coinbase -- it already has the entire L1 coinbase. And the whole block.

Namecoin works differently since Namecoin has the ability to survive if Bitcoin Core goes away. Bip301 chains, (BMM chains) however, do not.

So, the attack Peter describes is not possible.

False

5.2 Reorgs Invalidating Coins

The blind-merge-mine request has the following format:

- 3-bytes - Message header (0x00f00)
- 32-bytes - h* (side-Headerfoot)
- 1-byte - nSidechain (sidechain ID number)
- 4-bytes - prevInHeaderBytes (the last four bytes of the previous main: block)

The last field, prevInHeaderBytes, is designed to commit to the previous block header, ensuring that a blind-merge-mine request transaction is only valid in a specific block. This directly conflicts with an important principle of Bitcoin consensus design (reorgs should not invalidate transactions). For example, this is why coinbase outputs are only spendable after 100 blocks and why we have not and will not implement consensus-level transaction expiry.

As we all know, reorgs can allow for double spends. In that way, reorgs already allow for transactions to be invalidated.

Your node already handles a reorg, by disconnecting a block, and unwinding all the transactions there.

Ironically, shortly after Peter wrote this post, a serious LN vulnerability was disclosed. It involved fee-based-replacement of outputs -- Peter is a prominent champion of Replace-By-Fee. This led Peter to quickly and desperately push "OP Expire" - an op code for expiring outputs. This op code would require a softfork, and repurpose an unused OP NOP... exactly the same situation as OP Drivechain.

I strongly disagree that this was Satoshi's reason for the 100 block maturity period. I believe it was added to keep miners invested in the continual health of the blockchain. But this is a minor point.

False

6 The Lack of BIP-301 and BIP-300 Interaction

A very notable omission in BIP-301 is that BIP-301 does not interact with BIP-300 Hashrate Escrows. There is no mechanism to pay miners to ACK or NACK a specific BIP-300 hashrate escrow withdrawal proposal. The only way that coins can be withdrawn from a drivechain is by the direct participation of at least a majority of hash power over the 3 to 6 month withdrawal period.

BIP-301 clearly states in the Motivation section that it is intended to fix drawbacks of traditional merge

Bip300 proudly ignores what the L2 full nodes are doing -- to do otherwise would be to destroy the sidechain idea completely. So, Bip300 cannot force a sidechain to be written to be compatible with Bip301. Similarly it is possible to use Bip301 to mine an Altcoin -- similar to how vanilla Merged Mining is used to mine Namecoin. (Ruben Somers/SuperTestnet have already used BMM to mine other chains on BTC, today.)

BALTIC
HONEYBADGER
2015



Drivechain (BIPs 300+301)

WHEN DEVS COMPETE, USERS WIN

"DRIVECHAIN ... ARGUABLY COULD HAVE BEEN MORE IMPORTANT OR USEFUL THAN TAPROOT."

- Adam Back, Baltic Honeybadger 2022

LEARN via YouTube – **DOWNLOAD our Software** – Dev Notes

PEER-TO-PEER BITCOIN SIDECHAINS

Drivechain allows Bitcoin to *create, delete, send BTC to, and receive BTC from* “Layer-2”s called “sidechains”. Sidechains are Altcoins that lack a native “coin” – instead, pre-existing coins [from a different blockchain] must first be sent over.

“Sidechains” boils down to allowing consenting individuals to:

1. choose their own security models
2. spend their bitcoin how they like
3. permissionlessly create voluntarist technology

Once on a sidechain, coins can change hands an unlimited number of times, and in an

LINKS

- [Home](#)
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- [Friends of Drive](#)
- [Misinformation](#)
- [Telegram](#)
- [Twitter](#)
- [Reddit](#)
- [Sidechain Project](#)
- [Truthcoin.Info](#)
- [Bitcoin Hivemind](#)

- Deposit validation mempool rules refactored

OCT 18, 2020 (V35.00)

- Windows binaries!
- Automatically abandon failed BMM requests periodically (GUI version only)
- Add error checking for sidechain .dat files during init
- Updated testchain configuration file generator
- Fixed bugs related to sidechain deposits when there are multiple active sidechains.
- Bitcoin Core patch <https://github.com/bitcoin/bitcoin/pull/13622> to remove network request counting from wallet txns – remove DoS vector
- Fixes & improvements to BMM transaction abandonment code
- Fix reindex init bug: ignore sidechain number during AcceptBlock if we are reading from disk and reindexing
- Removed unused code from sidechain database & refactoring

SEPT 25ND, 2020 (V34.01)

- Withdrawals
 - Withdrawls can now be canceled. Use RPC ‘createtwrefundrequest’, or right-click from the GUI.
- If a withdrawal bundle (WT^A) fails there will be a waiting period before the next WT^A – use this time to cancel a withdrawal.
- Withdrawal bundles (WT^As) expire early, when they fall so far behind they can no longer succeed. This allows a new withdrawal bundle to be created ASAP.
- GUI
 - BMM table overhaul – the BMM table has been upgraded to include a lot more information.



8:10:41 / 8:25:04



Critics vs Supporters

Adam Back, Robin Linus, Rene Pickhardt, Fiatjaf, Anton Kumaigorodski, Super Testnet, Ben Arc, Christopher Burger, Junseth, Hampus Sjoberg, Michael Tidwell, David Bailey, Dan Held, Vlad Costea, Torkel Rogstad, Pete Rizzo, Bruce Fenton, Edan Yago, Guy Swann, Jameson Lopp, John Light, Andrew Bailey, Troy Cross, Brad Mills, Eric Wall, Nic Carter, Trevor.btc, Ari Paul, Joel Kaartinen, Alex Kravets, Timoleon Moraitis, Moonsettler, Sergio Demian Lerner, Marshall Long, Ruben Somsen, Charlie Shrem, Aaron van Wirdum ... etc etc

Remainder of Talk

1. Compare our ideas to the current state-of-the-art
2. Why non-mined L2s are doomed
3. “Soft forks without a soft fork” – technical accomplishments without a pull request to Bitcoin Core

All the World's Txns

03 May 2024

Txn Growth Rate				# of L2 chains	Blocks Per Year	Bytes per Txn	
6.20%				13	52596	138	
Year	World Txns/Year	BTC Adoption (% of world txns)	On-chain Txns/Year	Txns per chain	Txns per block	Imputed Blocksize Limit (megabytes per 10min)	6 Month Chain-History Size (terabytes)
2024	6.40E+12	1.00%	6.40E+10	4.92E+09	9.36E+04	12.92	0.340
2025	6.80E+12	2.00%	1.36E+11	1.05E+10	1.99E+05	27.44	0.722
2026	7.22E+12	5.00%	3.61E+11	2.78E+10	5.28E+05	72.84	1.916
2027	7.67E+12	10.00%	7.67E+11	5.90E+10	1.12E+06	154.72	4.069
2028	8.14E+12	20.00%	1.63E+12	1.25E+11	2.38E+06	328.62	8.642
2029	8.65E+12	50.00%	4.32E+12	3.33E+11	6.32E+06	872.48	22.944
2030	9.18E+12	100.00%	9.18E+12	7.06E+11	1.34E+07	1853.15	48.734
2031	9.75E+12	100.00%	9.75E+12	7.50E+11	1.43E+07	1968.04	51.756
2032	1.04E+13	100.00%	1.04E+13	7.97E+11	1.51E+07	2090.06	54.964
2033	1.10E+13	100.00%	1.10E+13	8.46E+11	1.61E+07	2219.65	58.372
2034	1.17E+13	100.00%	1.17E+13	8.98E+11	1.71E+07	2357.26	61.991
					12.9 MB (2025) → 2.3 GB (2035)		

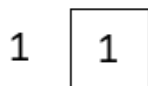
My Proposal 13-13

12.9 MB (2025) → 2.3 GB (2035)

Sidechains for Scaling -- Thunder Network

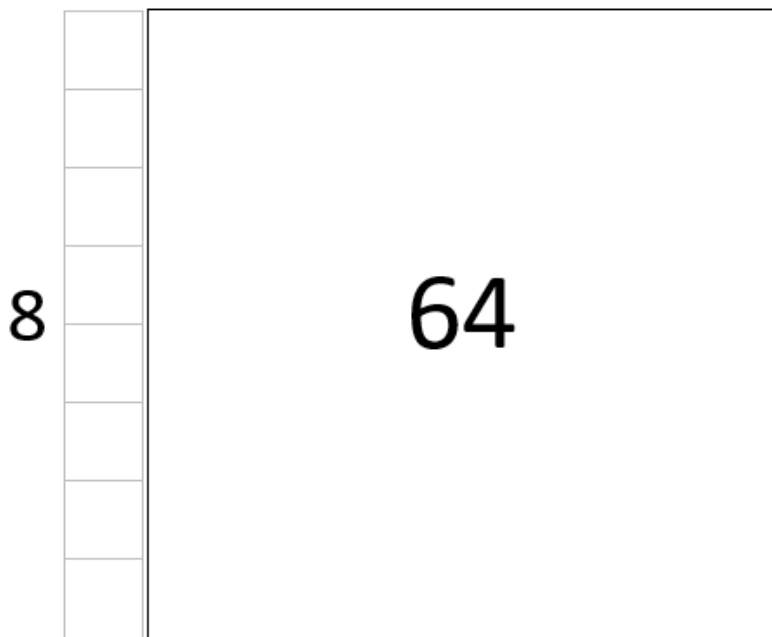
05 Feb 2021

Smallblock Chain



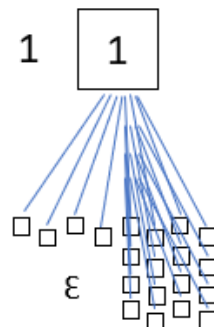
Throughput: 1
 $O(n^2)$ Cost: 1

Largeblock Chain
(No Sidechains)



Throughput: 8
 $O(n^2)$ Cost: 64

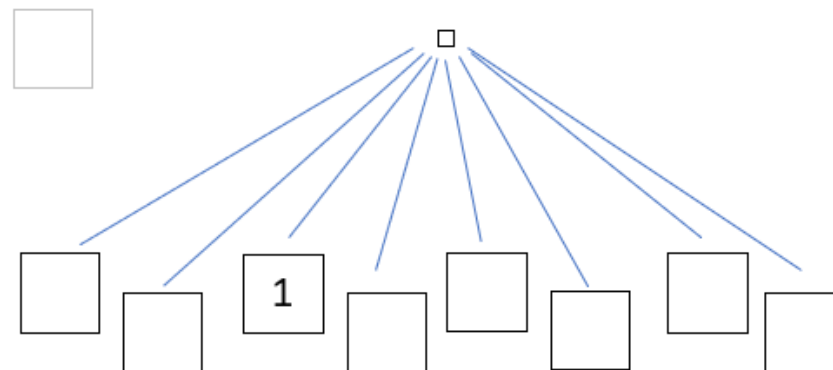
Lightning Network



Many ϵ (other people's L txns) that you disregard

Throughput: 8
 $O(n^2)$ Cost: $1+\epsilon$

Thunder Network



A total of 8 sidechains (each 1MB), 7 of which you disregard.

Throughput: 8
 $O(n^2)$ Cost: $1+\epsilon$

	BTC Cutting Edge (Lightning)	Our Better Strategy (Thunder 13-13) ¹
Onboarding	Cannot possibly onboard 8 billion people, ever. Fewer than 80 million people can use simultaneously. ²	Could onboard 8 billion people, over the next 5 years. ¹
Liquidity	Managing inbound liquidity is expensive, counterintuitive, and bad UX.	No such concept – (ie, always infinite liquidity).
Routes Failing	Has a failure rate of 5-25%. ³	Failure rate is 0%.
Payment Size	Larger payments (\$882+, 1M sats), have: higher fee, higher failure rate, longer txn time. ³	Larger payments treated just the same as smaller payments.
L1 Bytes	HTLCs consume too many bytes (L1 feerate dust effect), making them useless in practice. ¹	HTLCs not needed.
Fee-Pass Through	0% - (which will kill the security model, long term). ⁴	100% - (guaranteeing happy coexistence between drivechains and miners).
51% Attack	Miners can steal from individual LN channels, by censoring justice txns.	Stealing would take 3-6 months, be publicly transparent, and result in miners losing money. ⁵

¹ <https://www.truthcoin.info/blog/all-world-txns/#appendix-1--blockspace-napkin-math>

² <https://www.truthcoin.info/blog/lightning-limitations/>

³ https://fwc.widen.net/s/fxj6fgcwpg/fda_thelightingnetwork_expandingbitcoinusecases_1187503.1.0_v5

⁴ <https://www.youtube.com/watch?v=lmUCulfr1cE>

⁵ <https://www.drivechain.info/blog/fees/>

	BTC Cutting Edge (ARK)	Our Better Strategy (Thunder 13-13) ¹
Fee-Pass Through	0% - (which will kill the security model, long term). ²	100% - (guaranteeing happy coexistence between drivechains and miners)
Liquidity	Of all coins held by the ASP, only about 1/8 th can circulate. ³	There is no ASP, at all – and no liquidity constraint. Users deposit and use 100% of their own money.
Undefined	Still revised and changing, as of Jun 2024. ⁴	Nearly identical to original Nov 2015 design.
Note: [Rollups, Fedimint, Liquid, and Statechains] <i>also</i> do not pass fees to L1 miners.		

¹ <https://www.truthcoin.info/blog/all-world-txns/#appendix-1--blockspace-napkin-math>

¹ <https://www.youtube.com/watch?v=ImUCulfr1cE>

² <https://arkdev.info/blog/liquidity-requirements/#the-change-problem>

³ <https://thebitcoinmanual.com/articles/ark-v2/>

Category: Privacy

	BTC Cutting Edge (CoinJoin - Whirlpool 3x)	Our Better Strategy (DC-zSide-MeltCast)
Anonymity	Only mixes with 4 decoys per round – 124 maximum over 3 rounds. ¹	Mixes with every user of the system. No upper limit. Privacy increases with adoption.
Central Server	Must connect to a central “coordinator” server to use the service. ^{2 3}	No server, just a P2P network.
Interactive	The central server must maintain connection to the other 4 participants, and hope they don’t go AWOL. ^{2 3}	Not interactive at all – your computer can connect sporadically, whenever it wants, to the P2P network.
Mixable Values	Fixed mix values (100k, 1M, 5M, 50M sats). ¹	Can mix any value.
Extra Fee	Hardcoded fee (\$4 - \$ 1400) on top of the L1 blockchain fee. ¹	No extra fee – only normal (low) blockchain fee.
Complexity	Complicated to setup and use. ^{4 5}	Integrated into the L2 GUI – just click one button and wait.

¹ <https://bitcoiner.guide/whirlpool/>

² <https://github.com/Samourai-Wallet/Whirlpool?tab=readme-ov-file#using-whirlpool>

³ <https://bitcoinmagazine.com/culture/samourai-wallets-privacy-enhancing-whirlpool-now-in-public-beta>

⁴ <https://bitcoin-bumblebee.com/2023/08/19/whirlpool-coinjoins-on-sparrow-in-5-steps/>

⁵ <https://wiki.roninjo.io/en/setup/whirlpool>

Protecting Bitcoin's Properties

	BTC Cutting Edge (Naïve Ossification)	Our Better Strategy (DC + Precedence Ossification)
New Features	Approval process for new ideas is slow, and based on persuasion – the merge ultimately <u>changes L1</u> .	Bitcoiners are free to try any idea – on an L2 – whenever they like (ie, immediately). L1 is <u>unaffected</u> .
Ossification	Core currently releases a new version of Bitcoin every 6-7 months ¹ -- whether we want it or not. New CVEs are fixed secretly. ²	Innovation is pushed to L2s – so L1 can focus on remaining stable. L1 should then become simpler over time.
Developer Centralization	All our eggs are in one basket – we must hope that Core puts out a high-quality, competitive product – now and forever.	Developers <u>compete</u> to put out the best software. ³ Users switch among L1 + L2s, as needed. Bad software is left behind.
Disagreement	Dissatisfied BTC-users must jump ship to an Altcoin – even if they don't want to. ⁴ “Maybe you need a Monero”. ⁵	Dissatisfied users just make their own Bitcoin L2 – they remain Bitcoiners, but use different software.
Soft Forks	SegWit was a mandatory 4x Blocksize increase -- Taproot introduced yet another mangled address format.	Bip300 is an opt-in, ignorable, and reversible upgrade to Bitcoin Core that does not require changing any Core code.

¹ <https://bitcoincore.org/en/lifecycle/>

² <https://groups.google.com/g/bitcoinddev/c/Q2ZGit2wF7w?pli=1>

³ <https://www.drivechain.info/>

⁴ <https://www.drivechain.info/misinformation/#5>

⁵ <https://www.youtube.com/watch?v=ccJ33hLaMF0>

Mining

	BTC Cutting Edge (Stratum V2)	Our Better Strategy (EDAC Pools ¹)
Payout Threshold	Typically 0.001 BTC (\$109). (Miners must custody this with their pool.)	Can be 10x lower, (\$10).
Payout Frequency	Daily (for most, including Foundry). PPLNS would always need multiple blocks.	Can pay out every block (10 minutes).
Automatic Pool Switching	n/a – does not help	Can switch pools all day – thanks to PPS and fast payouts. This keeps pools honest.
Ease of Starting New Pools	n/a	Low pool switching costs = users can try your new pool, and get paid out quickly – instant credibility.
Job Negotiation	An unpopular, misconceived idea that no one wants – is contrary to logic, and to specialization-of-labor.	Unneeded. Instead, low-pool fee, high-FPPS among competitive pools, automatically deters txn censorship.
Pool Accountability	<u>Harms</u> accountability, by mudding the waters (re: job negotiation).	<u>Increases</u> accountability, by tracking all performance via apples-to-apples metrics (fee, and PPS rate).
Status	Proposed in 2019 (six years ago), in 2023 had only 1 dev, still not completed today, very low interest from real mining pools.	Will automatically happen, after Bip300 activates.

¹ <https://www.truthcoin.info/blog/ideal-mining/>

Fee Comparison

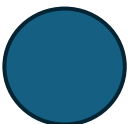
This diagram is to scale.



\$128 million
(fee revenue 2022)



\$767 million
(fee revenue 2023 –
the ordinals year)

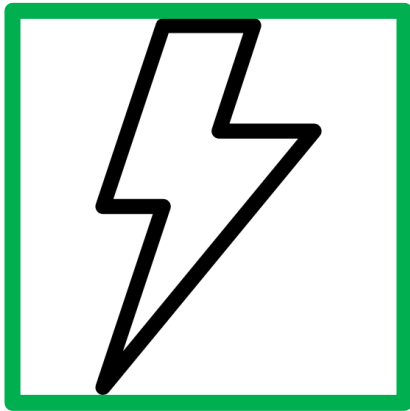


\$3.0 billion
(All-Altcoin Fee
Revenue 2022)

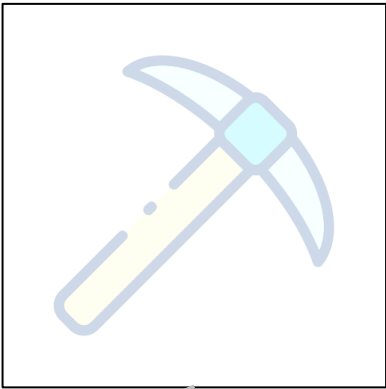


\$640 billion
(Earth's 6.4 trillion txn
X \$0.10 , 2022)

Non-Mined L2s VS Mined L2s



Non-Mined L2s



Mined L2s

L1



\$640 billion

The Coming L2 War

- *Instability*

1. Conflict between L1 and L2
2. Incentives for Mining Centralization
3. Vertical integration – eg “Foundry Lighting – AntPool LSP”
4. Original L2 security model obsolete

L2 Revenue

○ L1 Revenue

Every L2 is vulnerable to 51% attack by miners

[Lightning-dev] New form of 51% attack via lightning's revocation system possible?

René Pickhardt [r.pickhardt at gmail.com](mailto:r.pickhardt@gmail.com)

Tue Mar 13 13:30:21 UTC 2018

Since according to the spec channels should never be balanced worse than 99% to 1% the attacker could steal up to 99% of all the bitcoins allocated in the sum of all payment channels the attacker was connected to. This amount could obviously be way higher than just double spending her own funds. This attack would be interesting in particular for the power nodes created by the Barabasi-Albert model of lnd's autopilot (c.f.: <https://github.com/lightningnetwork/lnd/issues/677>).

I understand that with the growth of the bitcoin (mining) network a 51% attack becomes less and less likely. Also I am very happy to be proven false about the attack that I am describing.

Another sad thing about this attack is that I currently do not see any (reasonable) way of preventing this form of a 51% attack (other than creating payment channels that don't offer the possibility of revocation) as it is abusing exactly the core idea of lightning to do something in

onboard,
offboard,
and “justice”
txns on L1
can all be
censored

Centralization Advantage

- Lightning “LSP” backed by large hashrate
- Other LSPs cannot compete
- Eventually: all LSPs are backed by large pools

L2 Revenue
(Now MEV)

○ L1 Revenue

Non-Mined L2s
VS
Mined L2s



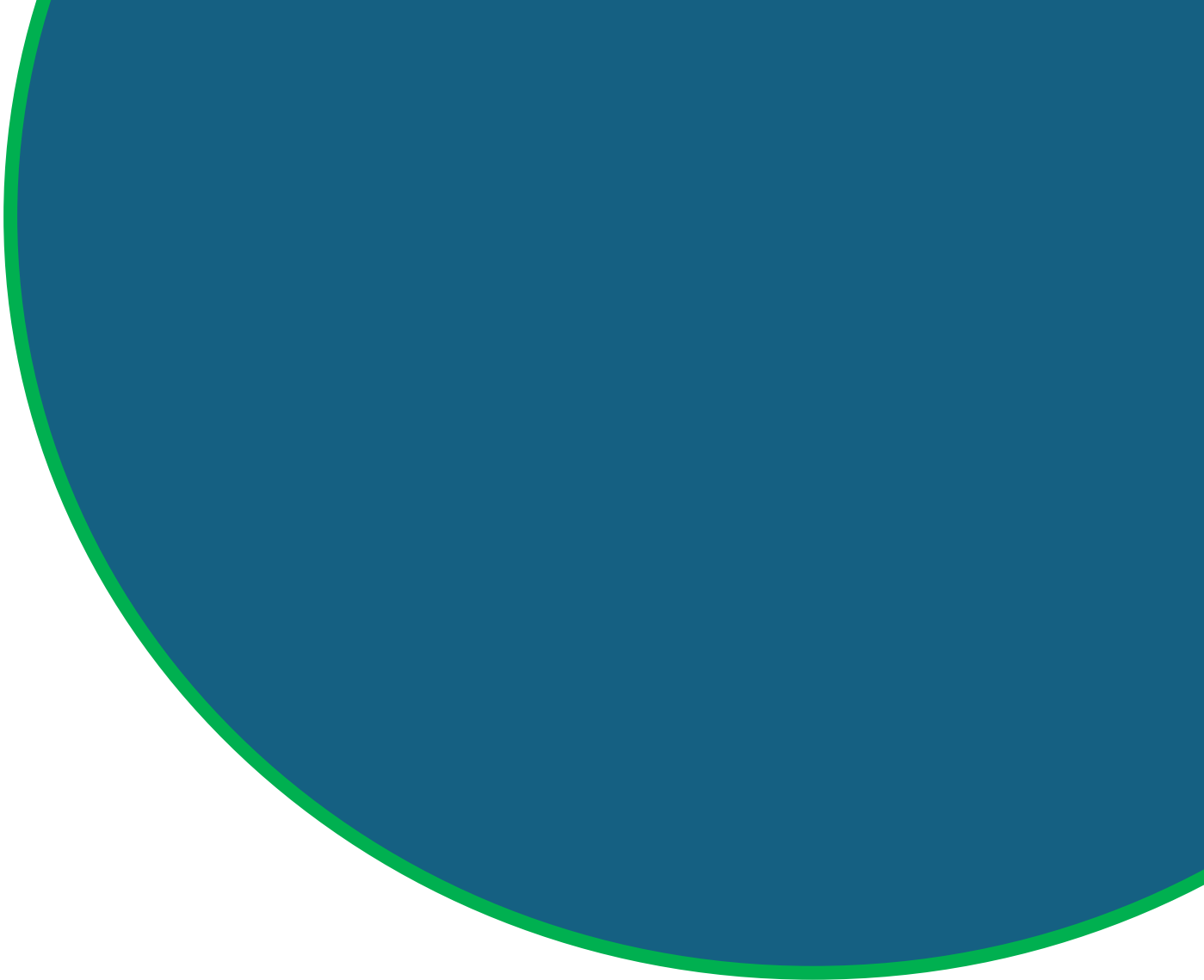
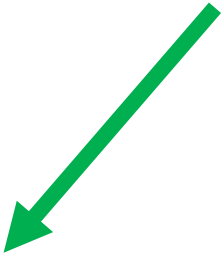
Non-Mined L2s



Mined L2s



L1



\$640 billion

CUSF

Core Untouched Soft Fork
“Ordinal-ization” of Soft Forks
“Soft Forks without a Soft Fork”

CUSF - “Core Untouched Soft Fork”
*or: “Soft Forks, without a Soft Fork”,
or “The Ordinal-ization of Soft Forks”*

Paul Sztorc
v0.4.1 -- 6/23/2024

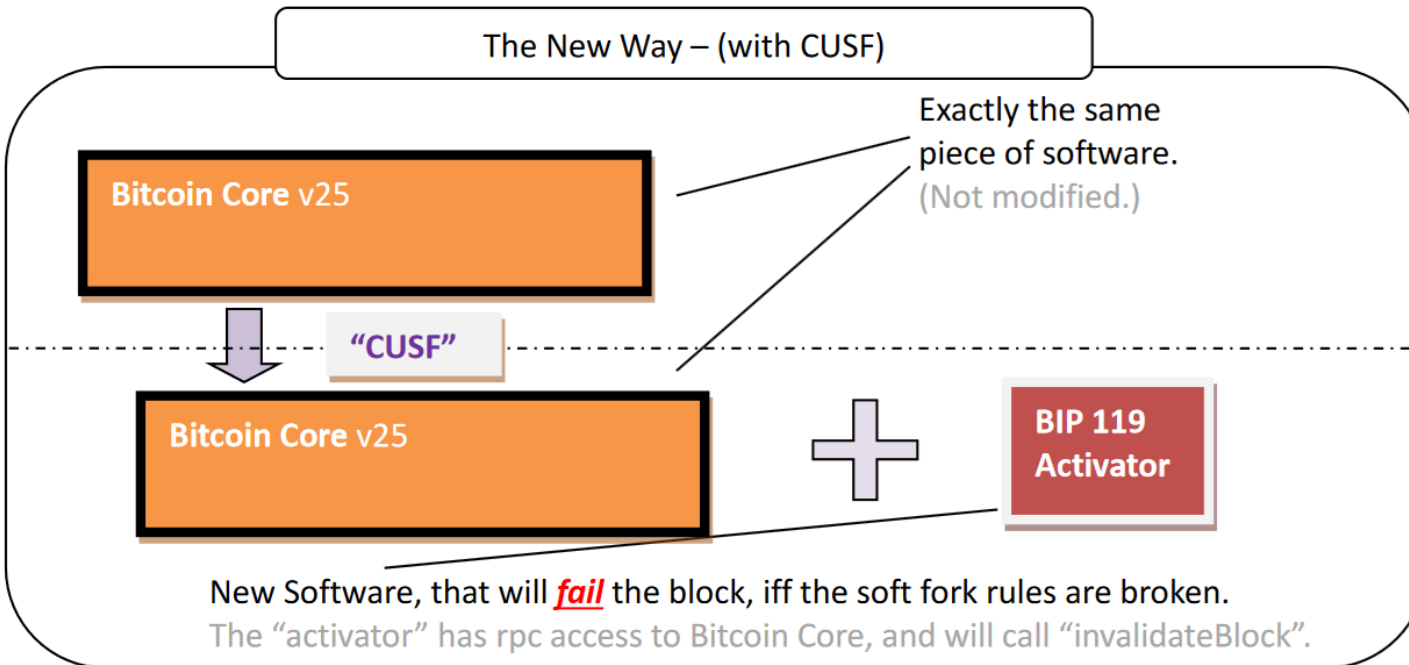
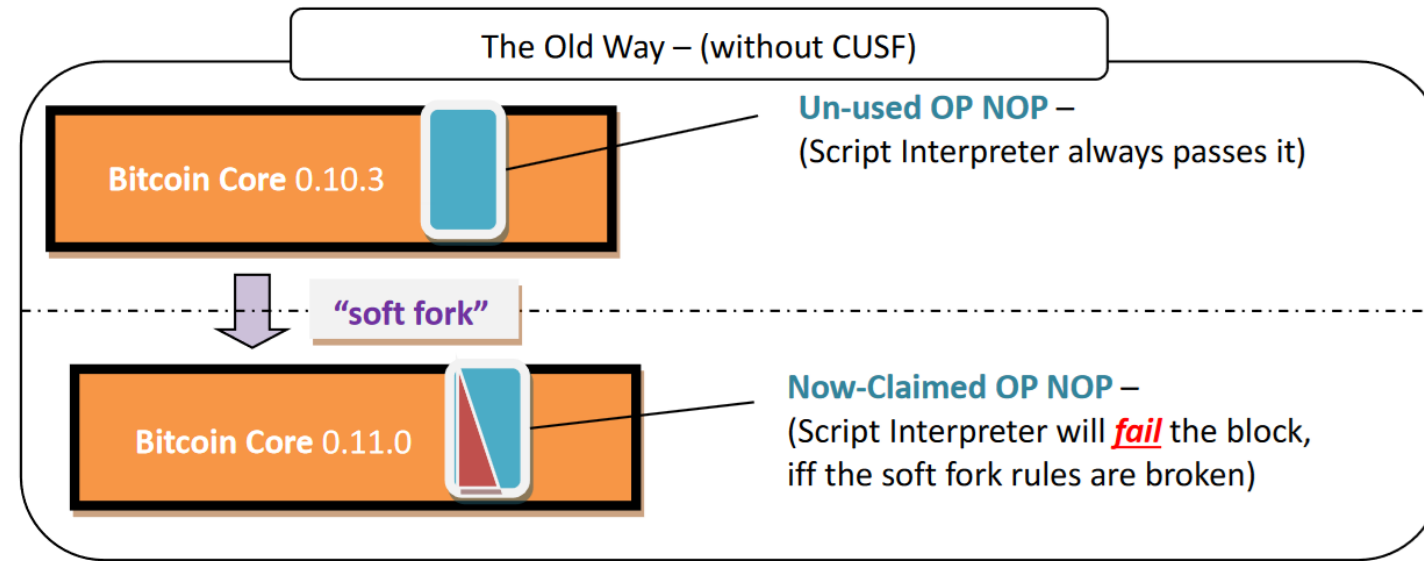
Summary

Each new soft fork (SF) should be a separate, standalone piece of software, “piloting” Bitcoin Core via the “invalidateblock” rpc. This makes soft forks faster, safer, and easier to understand -- ushering in a new age of Bitcoin Development.

The Idea

The current soft fork process is so vague that arguably no one knows what it is -- but it certainly involves opening a GitHub pull request. Here, I present an alternative process: put new soft fork validation rules in their own, separate piece(s) of software. This software can use “getblock” and “invalidateblock” (via rpc access to Bitcoin Core) to enforce new consensus rules. This has many advantages.

Paper at: <https://bip300cusf.com>



- Same effect
- But two software daemons
- Two RPC servers
- “Inefficient” for computer – 2x as much work
- But “separable” at the human level – socially scalable.
- Simple change, many advantages

Download Today – OP CAT & BIP300

Enforcers

Enforcers are standalone software meant to be used by people operating full nodes, to verify that their nodes are in compliance with the new soft fork rules to be implemented.

OP CAT 🐱

Download OP CAT Enforcer

Auditor

BIP300 🚗 🧑🏻

Download BIP300 Enforcer

Sidechain

Blockmaker

The blockmaker hacks getblocktemplate, to set a priority fee of -21M btc, for every txn rejected by any Enforcer. Therefore, miners who run this, will never have their blocks rejected by any Enforcer.

Download Blockmaker

bip300cusf.com/download

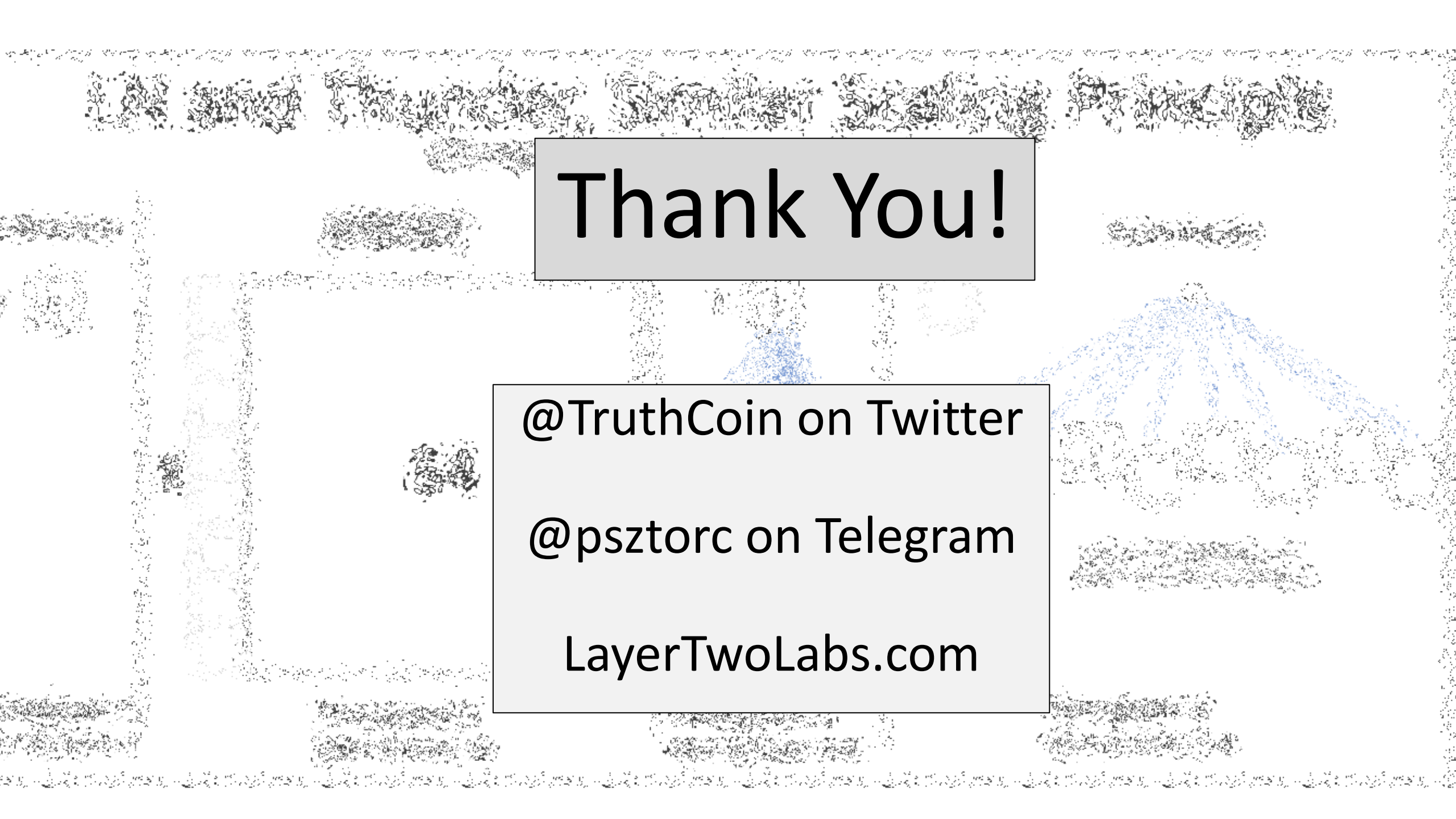
(This is one easy way of solving all of our soft fork problems.)

To Review: My Proposal Is...

1. Miners run the Bip300 CUSF activator,
(Bitcoin Core continues to do nothing).
2. Once 51% hashrate runs the CUSF addon, Bip300 activates.
3. Miners activate the Mined-L2s, (13-13) and collect \$\$ from them.

Problem solved!

(Bitcoin gets scalability & privacy & other things ..
... extensibility & security budget & ossification)



Thank You!

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@psztorc on Telegram

LayerTwoLabs.com